SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

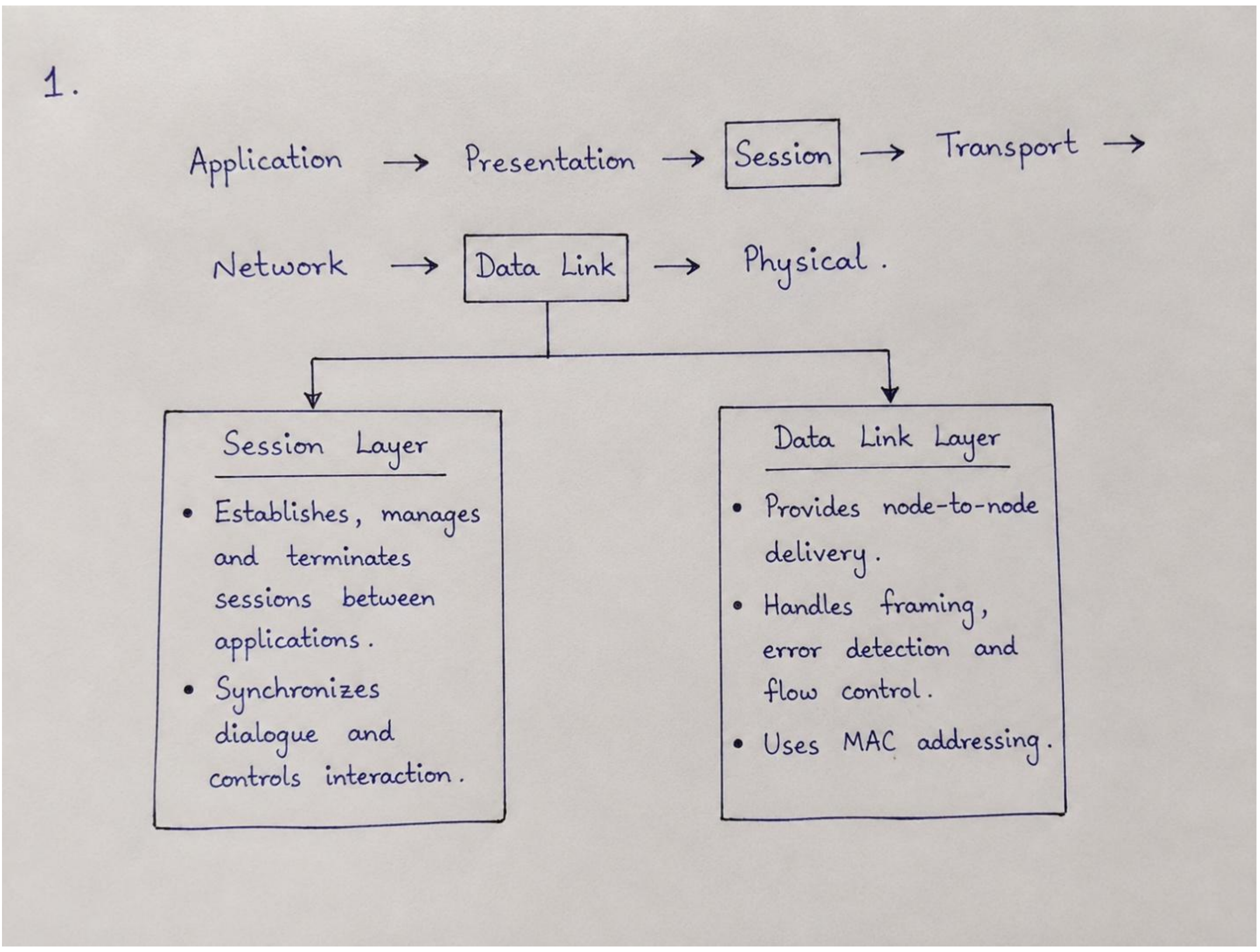
CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → ______ → Transport → Network → ______ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

Sol :

The missing layers in the OSI model are the Session Layer and the Data Link Layer.
The Session Layer establishes, manages, synchronizes, and terminates communication sessions between applications. It ensures that communication remains organized and allows interrupted sessions to be resumed when possible.
The Data Link Layer provides reliable node-to-node communication over a physical link. It performs framing, error detection, flow control, and MAC addressing to ensure that data reaches the correct device within the same network.
Together, these layers improve communication reliability. The Session Layer manages communication between applications, while the Data Link Layer ensures accurate data transfer over the local network before forwarding it to higher layers.



2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

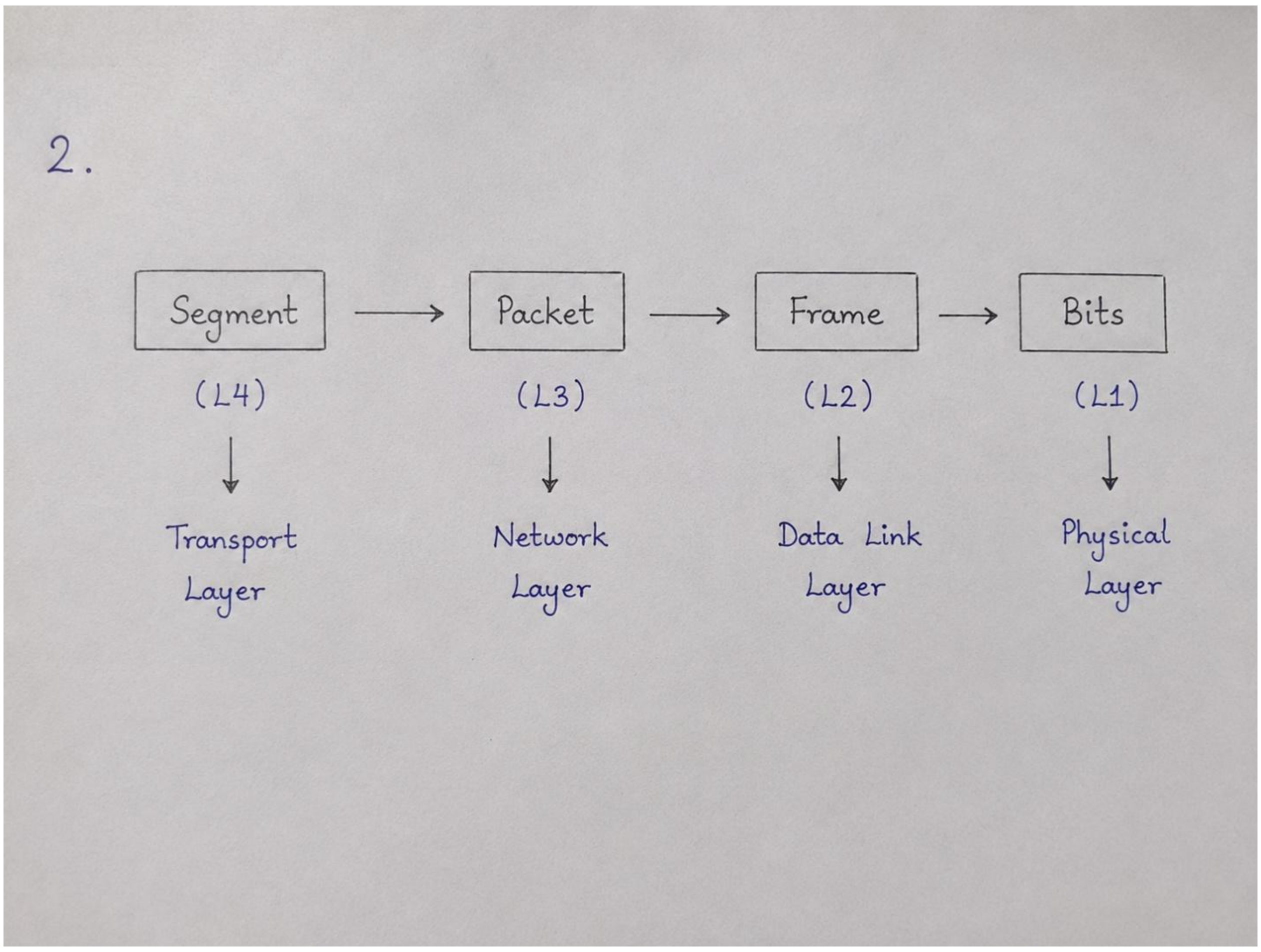
Sol:

As data travels from the sender to the receiver, it is encapsulated into different data units at each OSI layer.

- **Transport Layer → Segment**
- **Network Layer → Packet**
- **Data Link Layer → Frame**
- **Physical Layer → Bits**

Each layer adds its own control information to support reliable communication. The Transport Layer provides end-to-end communication, the Network Layer handles routing and logical addressing, the Data Link Layer performs framing and error detection, and the Physical Layer converts data into electrical, optical, or radio signals.

This transformation ensures reliable and efficient data transmission across networks.



3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

Sol:

Step-by-Step Explanation:

1. Data Link Layer → MAC Address

The Data Link Layer uses MAC addresses to identify devices within the same local network.

A MAC address uniquely identifies a network interface.

2. MAC Address → Local Delivery

MAC addresses are used to deliver frames from one device to another within the same Local Area Network (LAN).

Switches use MAC addresses to forward frames to the correct device.

3.Network Layer → IP Address

When the destination is on another network, the Network Layer uses IP addresses to identify the source and destination networks.

Routers examine IP addresses to determine the best path for the packet.

4. IP Address → Global Delivery

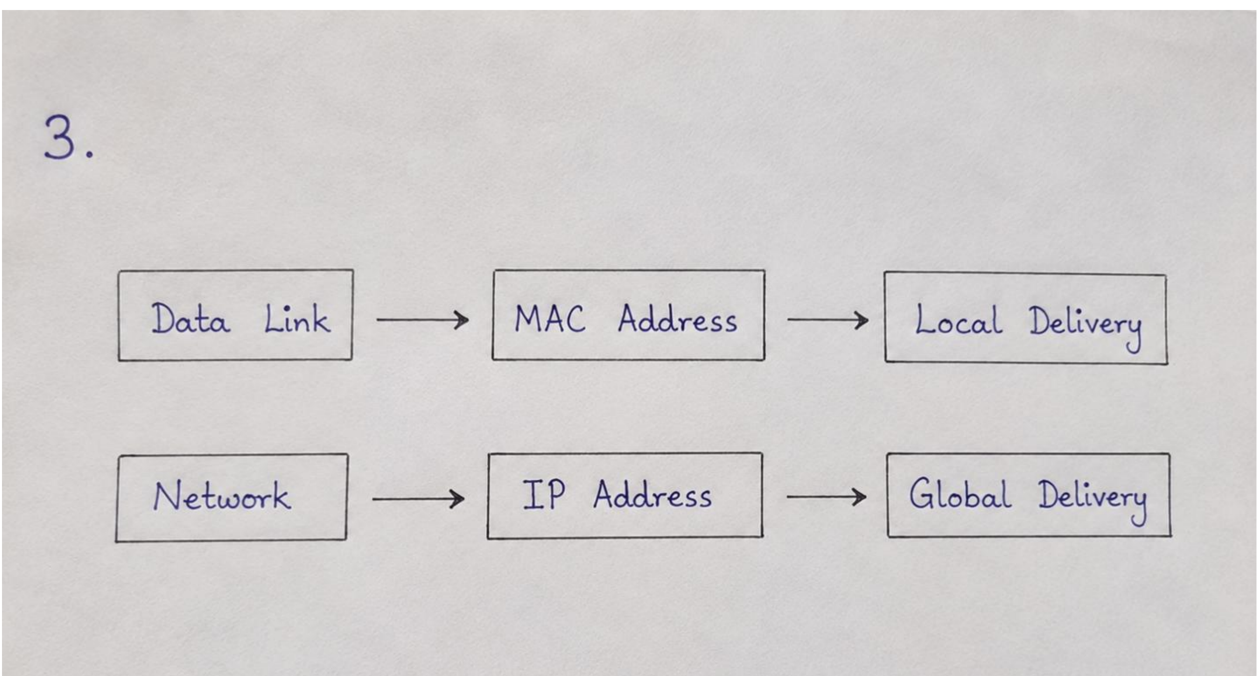
IP addresses enable packets to travel between different networks through routers. o This allows data to reach devices anywhere on interconnected networks, such as the

Internet.

Overall Relationship:

Data → IP Address (Global Routing) → Router → MAC Address (Local Delivery) →

Destination



4. A concept map shows:
ErrorControl
→DataLinkLayer→ErrorDetection
→ Transport Layer → Error Correction
Analyze how the interaction between these layers improves reliability in data transmission.
Sol:
Step-by-Step Explanation:
1. Error Control
Error control is used to identify and handle errors that may occur during data

transmission. It helps ensure that the data received is accurate and complete.

2. Data Link Layer → Error Detection

The Data Link Layer detects errors that occur during node-to-node communication.
It uses techniques such as CRC (Cyclic Redundancy Check) to check whether a frame has been corrupted.
If an error is detected, the damaged frame can be discarded or retransmitted.

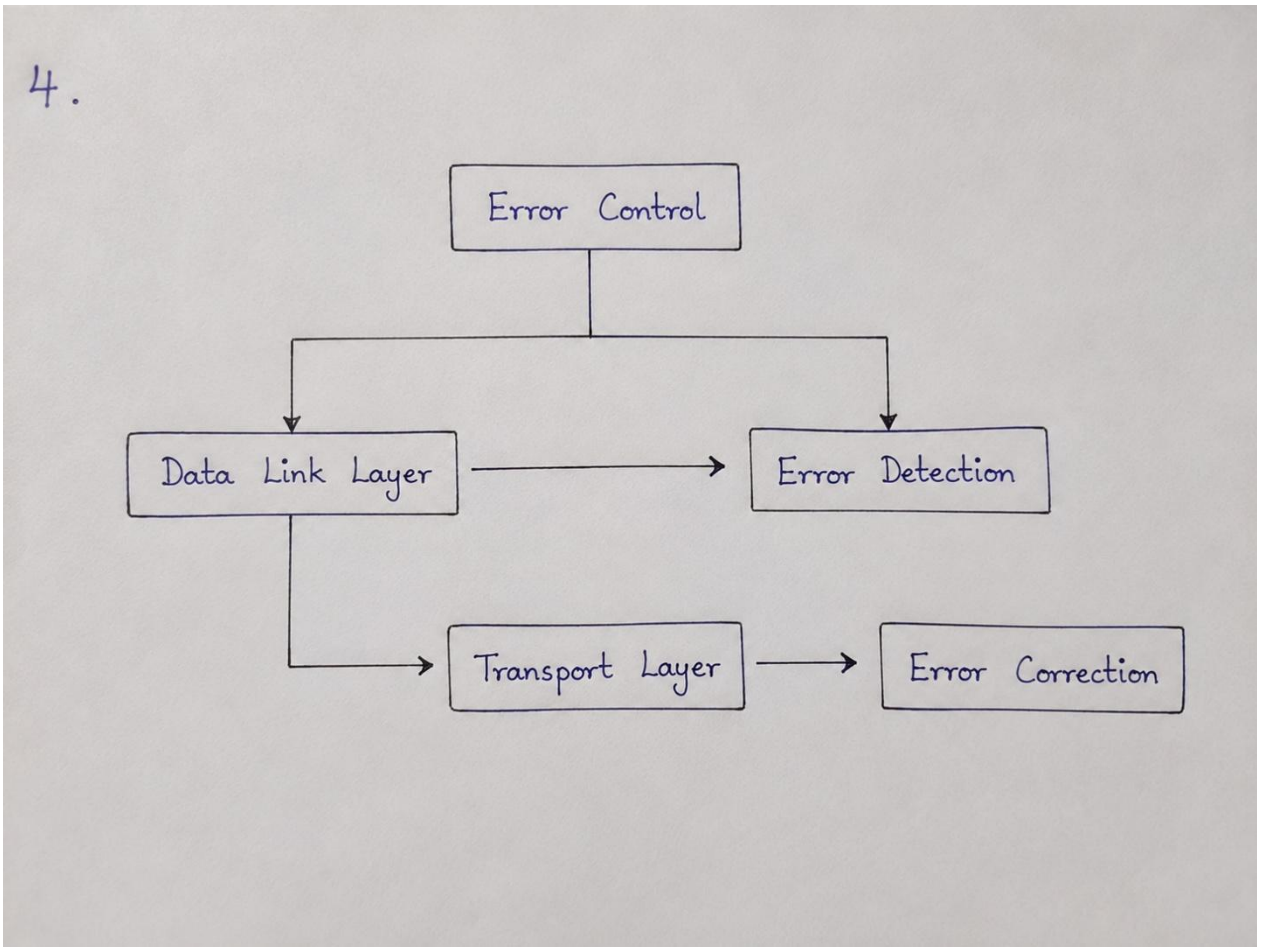
3. Transport Layer → Error Correction / Recovery

The Transport Layer provides end-to-end reliability.
Protocols such as TCP use acknowledgements, sequence numbers, and retransmission to ensure that missing or corrupted data is recovered.
It ensures that data arrives at the destination in the correct order and without loss.
Interaction Between Lavers:

Data Link Layer

→ Detects errors during local transmission
→ Prevents corrupted frames from moving forward

Transport Layer
→ Provides end-to-end error recovery
Retransmits lost or missing data
Ensures correct order and completeness



5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Sol:

Web Browsing Using OSI Layers

Step 1

The user enters a website URL in the browser.

Step 2

The Application Layer sends an HTTP/HTTPS request.

Step 3

The Presentation Layer encrypts and formats the data.

Step 4

The Session Layer creates the communication session.

Step 5

The Transport Layer provides reliable communication using TCP.

Step 6

The Network Layer selects the best path using IP addressing.

Step 7

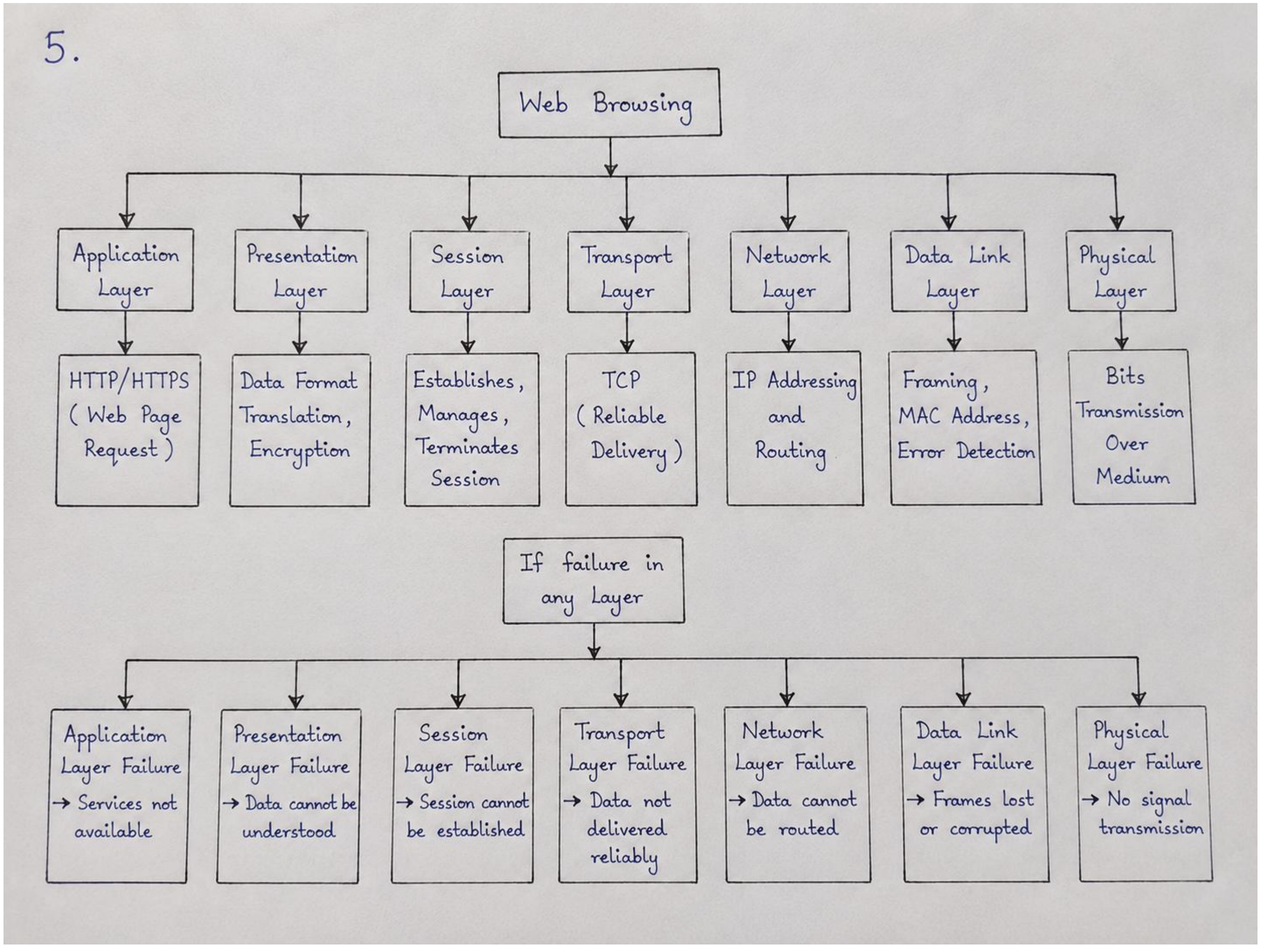
The Data Link Layer creates frames and uses MAC addresses.

Step 8

The Physical Layer converts frames into bits and transmits them.

Step 9

If any one layer fails, communication stops, because every layer depends on the previous layer.



6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✕ → Transport ✕ → Application ✕

Analyze the problem and explain how the concept map helps in troubleshooting.

Sol:

Problem Analysis:

1. Physical Layer

The physical connection is working properly.

Cables, Wi-Fi signals, and network hardware are functioning.

2. Data Link Laver

• Local network communication is working.

• MAC addressing and frame delivery are functioning correctly.

3. Network Layer - Main Problem

The problem is likely related to IP addressing, routing, or router configuration.

The device may have an incorrect IP address, subnet mask, default gateway, or routing issue.

Packets cannot reach the intended destination network.

4. Transport Layer

Since the Network Layer cannot deliver packets, a reliable TCP connection cannot be established.

Services using TCP may fail.

5. Application Layer

Applications such as web browsers, email, or other network services cannot communicate with remote servers.

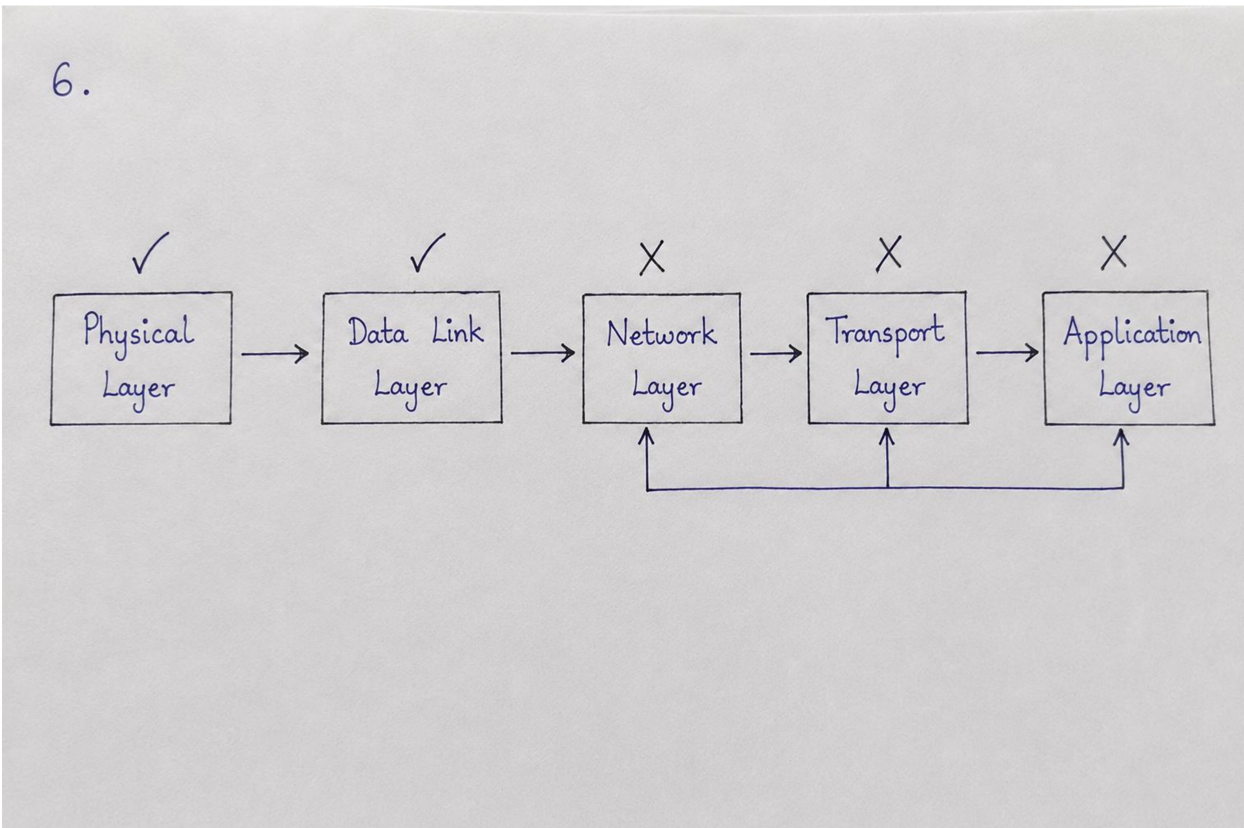
This is a result of the lower-layer failure, not necessarily the root cause.

How the Concept Map Helps in Troubleshooting:

The concept map provides a layer-by-layer troubleshooting approach. Since Physical and Data Link are already working, we can skip checking cables, Wi-Fi signals, switches, and MAC addresses.

The troubleshooting should focus on the Network Layer first

Check IP Configuration → Check Default Gateway → Test Router Connectivity→ Check Routing → Check DNS→ Test Application



RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation